

EVALUATING FEATURE PREDICTION FOR 2D POSE TRAJECTORY RESTORATION WITH PAIRED SYNTHETIC SUPERVISION

Anonymous authors

Paper under double-blind review

ABSTRACT

We examine whether predicting reference features improves the restoration of side-sensitive movement changes from noisy pose observations. This figure prototype checks scientific relationships and actual reading size in the official manuscript template.

1 CONTROLLED RESTORATION

States a and b share a source movement window. Physical mirroring changes both reference and rendered observation; naming corruption changes the observation alone. The model processes one state at inference. The mathematical schematic represents executed relationships and contains no synthetic example presented as a model result.

A States, references, and observations

$$M_a \xrightarrow{\text{knee edit } \mathcal{E}_\delta} M_b = \mathcal{E}_\delta(M_a)$$

$$Y_i = \pi_c(T_m M_i)$$

$$X_i = N_n[\text{Pose}(\text{Render}_{c,o}(T_m M_i))]$$

$i \in \{a, b\}$; paired camera / observation condition

T_m : physical 3D mirror; reproject Y_i

N_n : input naming error; Y_i unchanged

B Feature comparison during training

$$X_i \longrightarrow \text{student} + \text{predictor} \longrightarrow p_i$$

$$Y_i \longrightarrow \text{EMA teacher} \longrightarrow t_i$$

$$e_i = H[p_i/\tau_s - \text{sg}((t_i - c)/\tau_t)]$$

$$\text{Endpoint: } \|e_a\|^2 + \|e_b\|^2$$

$$\text{Delta: } \|e_b - e_a\|^2$$

Both add to the same base feature loss.

C Inference

$$X_i \longrightarrow \text{encoder} + \text{residual readout} \longrightarrow \hat{Y}_i$$

One observed state; no reference, paired state, teacher, or predictor.

Figure 1: Paired supervision with single-state inference. M_i is body-model motion, T_m the physical mirror, π_c the fixed-camera projection, and N_n an input naming transformation. Occlusion enters rendering. Feature residuals compare student predictions with privileged reference-teacher targets at common queries. The teacher is updated by exponential moving average of student weights; target gradients stop. Direct fitting instead updates the encoder and readout jointly.

2 DOES RESTORATION RECOVER MOVEMENT CHANGE?

Lower response error relative to observed coordinates does not establish useful change recovery. The zero-response benchmark is lower than every neural method in the pooled export. The next figure distinguishes jointly trained direct fitting from frozen-encoder readouts, and separates absolute scores from the paired effect of supervision.

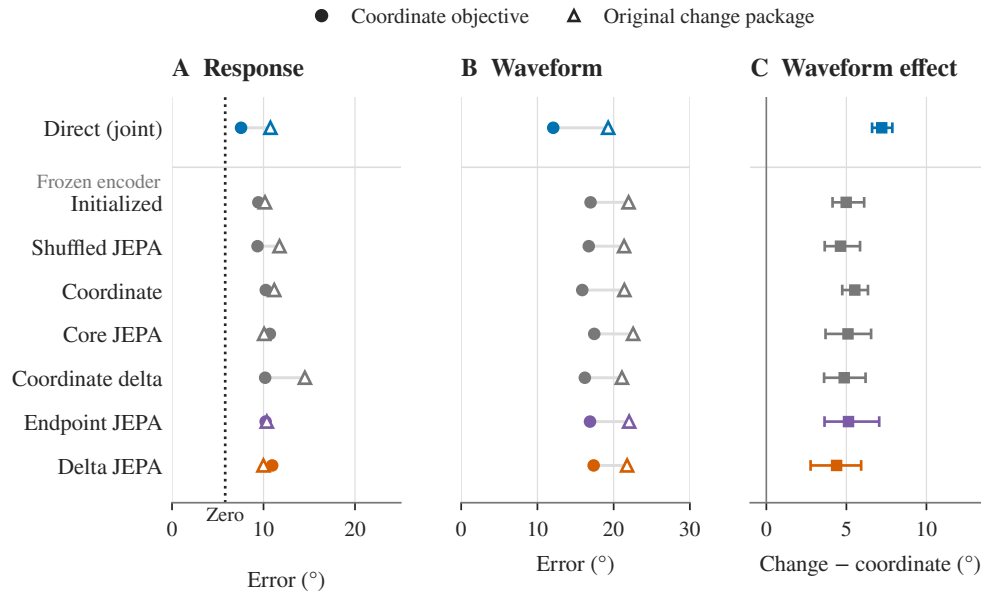


Figure 2: Restoration and supervision tradeoffs. A–B: descriptive hierarchical means over 14 development people and three fitted seeds, pooled over all estimators. Circles use coordinate supervision; open triangles add the original scalar-plus-geometry package. The zero-response reference has no waveform. C: change minus coordinate waveform error, with paired crossed-person/seed 95% bootstrap intervals (2,000 draws); positive values indicate deterioration. These are exploratory contrasts across correlated families. Scores retain failed measurements at 720° for response and 180° for waveform.